

4.1 Concept of Data Analysis

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔

CS XII

Chapter 4: Data and Analysis

Lecture 01 of 9

4.1 Concept of Data Analysis

LEARN IT

Data analysis is examining, organising and interpreting data so you can extract useful information. Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.

Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs. Five lines of Pandas often replace an hour of spreadsheet clicking.

WORKED EXAMPLE

A Hyderabad college has 400 CS marks in a spreadsheet. Analysis answers: What is the average? How many failed? Is the second shift weaker than the first? The numbers are data; those answers are information.

اردو میں سمجھیں

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔ پائٹھن اس لیے منتخب کیا جاتا ہے کہ وہ فائلیں، ڈیٹا بیس، حساب اور گراف سب سنبھال لیتا ہے۔

CHECK YOURSELF

1. Give one school use of data analysis.
2. Name two Python superpowers for analysis.

Answers: 1. Study results to improve teaching (or similar). 2. Any two: read CSV/Excel/databases, organise tables, statistics, charts.

Next lecture: 4.1.1 – 4.1.2 Data Sources and Database Connection